Macroeconomics in the Sequence Space

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April 13, 2026

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Sequence Space v.s State Space

Sequence Space v.s State Space

Motivation

In heterogeneous-agent models, the main difficulty is that the economy's state includes the whole cross-sectional distribution.
Sequence-space methods *avoid* carrying this object around explicitly when studying local dynamics around the steady state.

Definition 1

A state-space representation describes the economy recursively. At date t, a state variable x_t summarizes all payoff-relevant information from the past, and the model evolves according to

$$x_{t+1} = f(x_t, \varepsilon_{t+1}), \qquad y_t = g(x_t).$$

The key idea is: *once x_t is known, the future no longer depends on the past separately.*

Definition 2

A sequence-space representation treats equilibrium as a mapping between entire time paths of endogenous and exogenous variables. We solve for sequences $\{X_t\}_{t=0}^{\infty}$ such that the equilibrium conditions hold at all dates:

$$F\left(\{X_t\}_{t=0}^{\infty}, \{Z_t\}_{t=0}^{\infty}\right) = 0,$$

where X is the sequence of endogenous variables and Z is the sequence of shocks or policy paths.

Intuition:

- State space asks: *what is the economy's current state, and how does it transition?*
- Sequence space asks: *given a path of shocks, what equilibrium path of outcomes*

Sequence Space vs. State Space

Comparison

	State space	Sequence space
Unknowns	current state x_t	entire paths $\{X_t\}_{t=0}^{\infty}$
Core object	$x_{t+1} = f(x_t, \varepsilon_{t+1})$	$F\left(\{X_t\}_{t=0}^{\infty}, \{Z_t\}_{t=0}^{\infty}\right) = 0$
Linearization	around current state / transition law	around steady-state perfect-foresight paths
Main bottleneck	high-dimensional state, esp. the distribution	large Jacobians, but easy composition/inversion

Sequence Space v.s State Space

Takeaways

When the distribution is the hard state variable, sequence space often turns a "huge state" problem into a "structured Jacobian" problem.

SSJ Methods

SSJ development

SSJ in Continuous Time

To be developed

Determinacy and Large-Scale Solutions in the Sequence Space

To be developed

2nd SSJ

To be developed

fiscal and monetary policy

To be developed

behavioural macro

To be developed

life-cycle model

To be developed

dynamic spatial model

To be developed

production network

To be developed

het households + het firms

To be developed

het banks

To be developed

optimal policy

To be developed

macro labor

To be developed

Global Solution

Global Nonlinear Solutions in Sequence Space and the Generalized Transition Function

To be developed

DL in sequence space

To be developed

RL in sequence space

To be developed

Reference

Reference

To be developed